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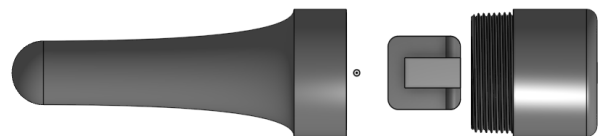
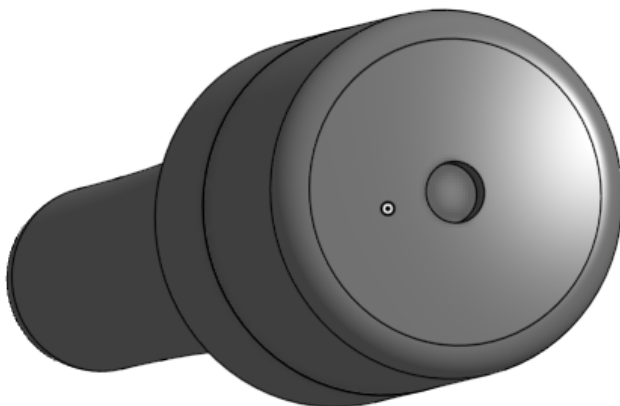
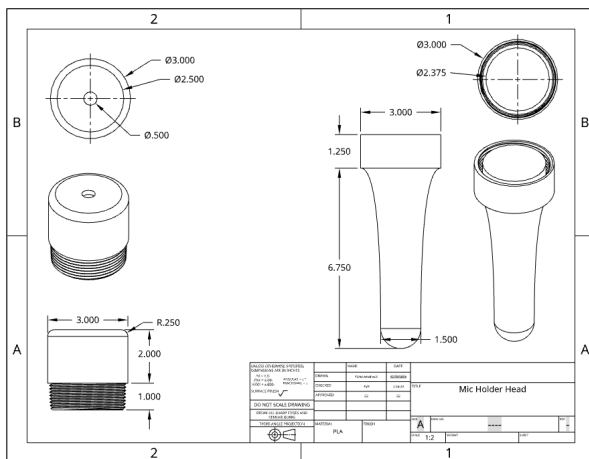
## Element G: Construction of a Testable Prototype

### Mic Holder prototype 1

Date: 2/28/24-3/1/24

Assembly drawing of Mic Holder 1

3D CAD models of first prototype

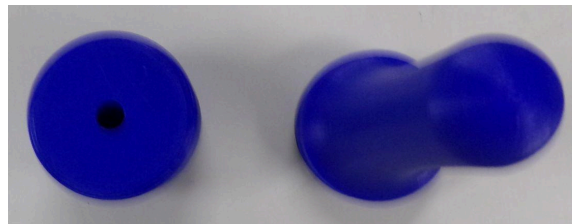
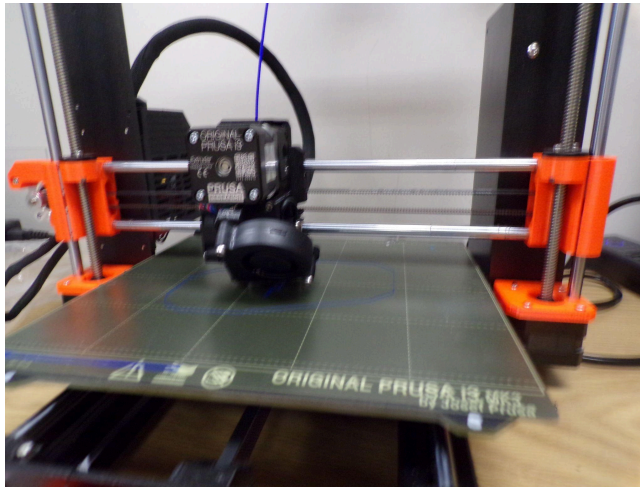


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3D Printed model



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Design 1 to 2: Printing of threads on the first design failed, so the head did not screw onto the holder. Also, the mic holder was way too large and did not look TV authentic

|        | Details                                                       | Tools and Manufacturing Methods used                                                                                      |
|--------|---------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------|
| Step 1 | Reflection and discussion of selected mic holder illustration | Paper to draw illustration on, marker to draw illustration, ruler to precisely draw lines                                 |
| Step 2 | Onshape draft of mic transmitter created                      | Onshape to create 3D model, PC to use Onshape program                                                                     |
| Step 3 | Onshape draft of mic head created                             | Onshape to create 3D model, PC to use Onshape program                                                                     |
| Step 4 | Onshape draft of mic holder created                           | Onshape to create 3D model, PC to use Onshape program                                                                     |
| Step 5 | Full mic holder assembly drafted                              | Onshape to create 3D model, PC to use Onshape program                                                                     |
| Step 6 | Mic holder assembly printed                                   | Onshape to create 3D model, PC to use Onshape program, Prusa slicer to prepare print, Prusa 3D printer to print prototype |

Notes of the progression from your blueprint to your (various) physical prototype(s), including why and how aspects of your design may have changed:

We changed the way the head secures, as at first it was a strap/button system, and after it was a screw on top. We also changed the size of the sound hole on the top. There was a printing error in which the cylinder in the center of the holder was misaligned, therefore restricting the head from screwing on. The screw threads would not work, so the prototype was a failure. Also, there was too much space in the transmitter holder slot, so the SM58

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transmitter was not held securely. More of an aesthetic flaw, the mic head looked way too large, and did not look like an authentic TV microphone.

## Mic holder prototype 2

Date: 3/4/24 - 3/8/24



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Date: \_\_\_\_\_

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3D printed model



Name: \_\_\_\_\_

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Class: \_\_\_\_\_

|        | Details                                                                                    | Tools and Manufacturing Methods used |
|--------|--------------------------------------------------------------------------------------------|--------------------------------------|
| Step 1 | Reflection on failed print (threads failing, center part printer misaligned, head too big) | Holder prototype, Onshape, PC        |
| Step 2 | Demonstration to stakeholder, feedback received, tests conducted                           | Holder Prototype, SM58 transmitter   |
| Step 3 | Measurements of transmitter recorded                                                       | Ruler, SM58 Mic Transmitter          |
| Step 4 | Modification of prototype; sizes changed                                                   | Onshape, PC                          |
| Step 5 | Printing of improved prototype                                                             | Prusa Software, Prusa Printer        |

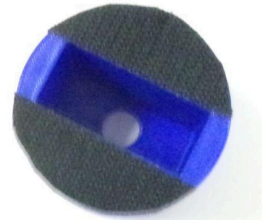
## Mic holder prototype 3

Design 2 to 3: Printing of threads failed again, and head did not screw on. Replaced the threads design with velcro to guarantee a secure fit.

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Name: \_\_\_\_\_

Date: \_\_\_\_\_

Class: \_\_\_\_\_

|        | Details                                                       | Tools and Manufacturing Methods used                                                         |
|--------|---------------------------------------------------------------|----------------------------------------------------------------------------------------------|
| Step 1 | Reflection and discussion of selected mic holder illustration | Paper to draw illustration on, marker to draw illustration, ruler to precisely draw lines    |
| Step 2 | Onshape draft of mic transmitter created                      | Onshape to create 3D model, PC to use Onshape program                                        |
| Step 3 | Onshape draft of mic head created                             | Onshape to create 3D model, PC to use Onshape program                                        |
| Step 4 | Onshape draft of mic holder created                           | Onshape to create 3D model, PC to use Onshape program                                        |
| Step 5 | Full mic holder assembly drafted                              | Onshape to create 3D model, PC to use Onshape program                                        |
| Step 6 | Mic holder assembly printed                                   | Onshape to create 3D model, PC to use Onshape program, Prusa 3D printer to printer prototype |
| Step 7 | Velcro applied                                                | 3D printer model, velcro tape                                                                |

## Mic holder prototype 4

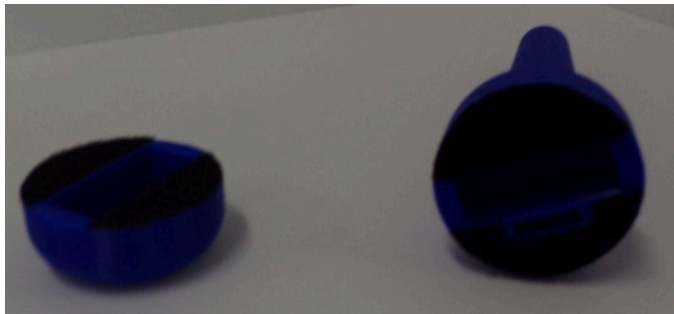




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Design 3 to 4: Mic head changed to a round shape to look more authentic. Reduces size of holder slightly. Changed design so that the transmitter does not just fit in the head: the gap is shared between the holder and the head so the transmitter stays securely in the middle.

|        |                                                               |                                                                                           |
|--------|---------------------------------------------------------------|-------------------------------------------------------------------------------------------|
| Step 1 | Reflection and discussion of selected mic holder illustration | Paper to draw illustration on, marker to draw illustration, ruler to precisely draw lines |
| Step 2 | Onshape draft of mic transmitter created                      | Onshape to create 3D model, PC to use Onshape program                                     |
| Step 3 | Onshape draft of mic head created                             | Onshape to create 3D model, PC to use Onshape program                                     |
| Step 4 | Onshape draft of mic holder created                           | Onshape to create 3D model, PC to use Onshape program                                     |
| Step 5 | Full mic holder assembly drafted                              | Onshape to create 3D model, PC to use Onshape program                                     |

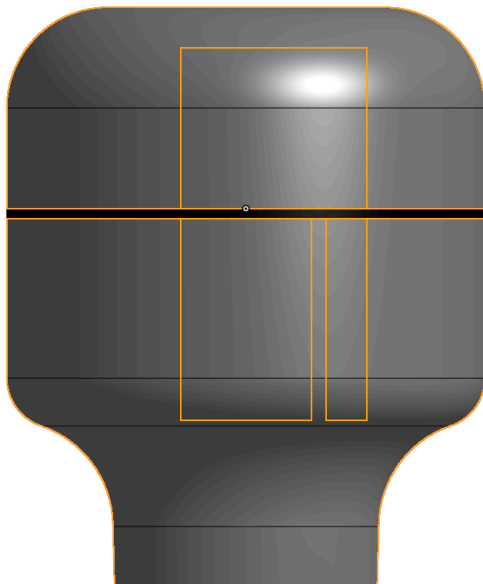
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|        |                             |                                                                                            |
|--------|-----------------------------|--------------------------------------------------------------------------------------------|
| Step 6 | Mic holder assembly printed | Onshape to create 3D model, PC to use Onshape program, Prusa 3D printer to print prototype |
| Step 7 | Velcro applied              | 3D printer model, velcro tape                                                              |

## Mic holder prototype 5



|        |                                                               |                                                                                           |
|--------|---------------------------------------------------------------|-------------------------------------------------------------------------------------------|
| Step 1 | Reflection and discussion of selected mic holder illustration | Paper to draw illustration on, marker to draw illustration, ruler to precisely draw lines |
|--------|---------------------------------------------------------------|-------------------------------------------------------------------------------------------|

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Class: \_\_\_\_\_

|        |                                           |                                                                                            |
|--------|-------------------------------------------|--------------------------------------------------------------------------------------------|
| Step 2 | Onshape draft of mic transmitter modified | Onshape to edit 3D model, PC to use Onshape program                                        |
| Step 3 | Mic holder assembly printed               | Onshape to create 3D model, PC to use Onshape program, Prusa 3D printer to print prototype |
| Step 4 | Velcro applied                            | 3D printer model, velcro tape                                                              |

Design 4 to 5: All we changed was the width of the slot for the transmitter, because in our previous design the holder was off center due to measurement issues. This drawing shows the adjusted size of the slot on the head piece and how it lines up with the slot on the bottom part.

Notes of the progression from your blueprint to your (various) physical prototype(s), including why and how aspects of your design may have changed:

Design 1 to 2: Printing of threads on the first design failed, so the head did not screw onto the holder. Also, the mic holder was way too large and did not look TV authentic

Design 2 to 3: Printing of threads failed again, and head did not screw on. Replaced the threads design with velcro to guarantee a secure fit.

Design 3 to 4: Mic head changed to a round shape to look more authentic. Reduces size of holder slightly. Changed design so that the transmitter does not just fit in the head: the gap is shared between the holder and the head so the transmitter stays securely in the middle.

Design 4 to 5: All we changed was the width of the slot for the transmitter, because in our previous design the holder was off center due to measurement issues. This drawing shows the adjusted size of the slot on the head piece and how it lines up with the slot on the bottom part.



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List of material details, including item name, description, where it was purchased/obtained (include URL if applicable), cost per unit (if applicable), # of units, total cost (if applicable)

| Material name     | Description                           | Where material was purchased                                                                                                                                                            | Cost per unit | # units | Total cost per item |
|-------------------|---------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|---------|---------------------|
| Prusa 3D printer* | Original Prusa i3 Mk3S + 3D printer   | Prusa<br><a href="https://www.prusa3d.com/en/product/original-prusa-i3-mk3s-3d-printer-3/">https://www.prusa3d.com/en/product/original-prusa-i3-mk3s-3d-printer-3/</a>                  | \$899         | 1       | \$899               |
| PolyTerra PLA     | Bioplastic based 3D printing filament | Polymaker<br><a href="https://us.polymaker.com/products/polyterra-pla?variant=39574351511609">https://us.polymaker.com/products/polyterra-pla?variant=39574351511609</a>                | \$20          | 1/10    | \$2                 |
| Onshape           | 3D CAD designing program              | Onshape<br><a href="https://www.onshape.com/en/pricing">https://www.onshape.com/en/pricing</a>                                                                                          | \$0           | 3       | \$0                 |
| Velcro            | Trazaon hook & loop adhesive tape     | Amazon<br><a href="https://www.amazon.com/Adhesive-Industrial-Strength-Strong-Strips/dp/B097TWHBTG">https://www.amazon.com/Adhesive-Industrial-Strength-Strong-Strips/dp/B097TWHBTG</a> | \$17.99       | 1       | \$17.99             |

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**\* PRUSA 3D PRINTER WAS PROVIDED BY SCHOOL, COST IS NOT APPLICABLE**

Detailed sketches, which might include CAD if you have learned it in your class:

**Construction of the Final Prototype (G2)**

When you've finished construction of your final prototype, you must be sure that your prototype allows sufficient functionality that you could collect objective, measurable data for all or nearly all of your design requirements. This requires you to look back at your design requirements from Element C. Copy all of your design requirements from Element C into this table and then add how your prototype will allow you to collect objective, measurable data. Note any weaknesses you find or details that will require expert review because they cannot be mathematically modeled or tested,

For example, if you have a design requirement that your design can "transport medicine a distance of 5 meters in 1 minute", is your prototype sufficiently functional such that you can test that? (e.g. can the prototype move the medicine at least a little bit?)

| Design Requirement                   | How will this prototype allow you to collect objective, measurable data?                                                                                                                                                                       |
|--------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Can firmly hold SM58 Mic Transmitter | This prototype will allow our group to collect objective data because we can insert the mic transmitter and measure whether or not the transmitter will fit inside the slot. We can measure the dimensions of the slot compared to the holder. |



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|                                               |                                                                                                                                                                                                                                                                                          |
|-----------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Looks like an authentic television microphone | This prototype will allow our group to collect data because we can compare it to previous renditions of the microphone, and to real, authentic microphones; it would be easy to determine whether or not the microphone looks authentic only using common sense and deductive reasoning. |
| Comfortable to hold                           | Our group can have multiple people test the grip, and they can give feedback based on the comfort of the shaft. Their feedback will allow us to collect data.                                                                                                                            |
| Weighs less than 2 pounds                     | We can simply weigh the entire microphone holder assembly on a scale to determine the weight. We would be able to collect measurable and objective data.                                                                                                                                 |
| Less than 3x3x15 inches in volume             | We can measure the volume of the microphone assembly by measuring the assembly or using volume equations. This data is measurable and objective.                                                                                                                                         |
| Able to have a mic flag fit over it           | Our group can collaborate with the mic flag group to determine whether or not our microphone fits in their flag. We would be able to collect objective data by simply determining whether or not the flag fits around the holder and looks good.                                         |
| Durable material                              | Our group can determine the durability of the holder by testing whether or not the holder will break due to falls, blunt force, or any other type of impact.                                                                                                                             |

### Testing or Modeling of Attributes (G3, G4) and Addressing Non-Testable Attributes (G4)

Describe all of your prototype's attributes (features, aspects, details, or subsystems). Demonstrate that your prototype is constructed with sufficient functionality that your prototype can test all attributes of your design. This requires you to look back at your initial proposed design details from Element D.



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For example, if your initial design details called for wheels and motors, is your prototype sufficiently functional such that you can test out those wheels and motors?

If there are aspects of your design that you believe cannot be physically prototyped or mathematically modeled within the scope and limits of your class, you must later provide evidence that an expert has reviewed your justifications and that you have incorporated their feedback. You do not need to conduct that review in Element G. For now, enter “expert review required” for these attributes in the final column.

| Attribute                           | Description                                                                                                                                                                    | Describe your prototype's functionality to test this attribute                                                                                                                                                                          |
|-------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Head able to be removed from holder | The head of the assembly is able to disconnect from the holder sufficiently. The transmitter is able to fit inside the slot and not fall/break during the head removal process | Our prototype uses velcro to attach/detach the head from the holder. The slot being shared between the holder and the head means that it will stay in the holder during the entire process.                                             |
| Sound is clear                      | The sound picked up by the SM58 mic transmitter inside the holder assembly is clear. The head does not disturb the sound received.                                             | Our prototype features a hole on the top-center of the head, right above where the transmitter picks up sound. We are able to prove the sound is clear by speaking into the microphone and testing the sound emitted from the receiver. |



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Class: \_\_\_\_\_

**Addressing Non-Testable Attributes (G4)**

For all attributes that you listed "expert review required" in the table above, provide a justification for why that is true. "Field experts" are professionals who spend at least a portion of their time working in a field directly related to your project. These field experts are not permitted to double count as stakeholders.

| Attribute    | Justification for why this attribute cannot be modeled or tested with your prototype                                                                                                 |
|--------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Authenticity | Authenticity is a perceived sense not a numerical quantity. Field experts will be able to review the microphone by comparing it to real microphones and seeing if they look similar. |

